

University of Moratuwa
IEEE Student Branch

A Student Chapter of IEEE Signal Processing Society



Low-Light Image Denoising & Enhancement Challenge

Official Document of the Competition Details

1. Introduction

Welcome to the Low-Light Image Denoising Challenge!

Capturing clear, high-quality photos under low-light conditions is a major challenge for modern digital photography. When a camera operates in a dark environment, the limitations of the camera sensor become more noticeable, resulting in read noise, thermal noise, unwanted grain, and visual distortion in the captured image.

This competition invites undergraduate students to tackle this real-world problem. Participants will work with a custom dataset prepared from clean mobile-phone photographs and corrupted using a synthetic low-light noise model that reproduces several common sensor-related artifacts. The goal is to design effective denoising algorithms capable of removing these complex low-light artifacts while preserving image details, structure, and visual quality.

Whether you are exploring image processing for the first time or testing novel algorithmic approaches, this challenge provides a hands-on opportunity to solve an authentic digital imaging problem. Put your problem-solving skills to the test, clean up the corrupted images, and gain practical experience with real-world digital signal processing.

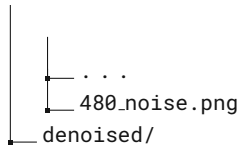
2. Competition Details

2.1 The Task Details

- The challenge for each team is to denoise a collection of noisy images.
 - **The dataset:**
https://drive.google.com/drive/folders/1kxh5m5Rm_3EB9oNK0nfH0MeTqXx9zyNq?usp=drive_link
 - The dataset contains different severity levels of noise.
 - Images with a resolution of 992×992 pixels are provided.
 - The complete dataset consists of 500 noisy images and 500 ground-truth images.
 - A dataset of 460 noisy images (001_noise.png–460_noise.png) and their ground-truth images (001.png–460.png) will be provided.
 - You will also receive 20 noisy images (461_noise.png–480_noise.png) without their corresponding ground-truth images. You must denoise these images and include the 20 denoised images in the preliminary-round submission.
 - A further 20 image pairs (images 481–500) will be kept fully hidden by the organizers for the physical final round. Finalists will receive only the corresponding noisy images during the final round.
 - The provided noisy images must not be renamed.
 - The codebase and directory instructions are available in the following GitHub repository:
https://github.com/moraspcup-official/mora_sp_cup_2026.git
- The repository is organised into four main directories:
1. The baseline implementation (baseline/).
 2. The evaluation scripts (evaluation/).
 3. The competition dataset files (competition_data/).
 4. The Python scripts and other files created or used by teams (scripts/).

To see how the baseline works, teams must place the images in the correct directories as shown below.

```
competition_data/
├── public/
│   ├── ground_truth/
│   │   ├── 001.png
│   │   ├── ...
│   │   └── 460.png
│   └── noisy/
│       ├── 001_noise.png
│       ├── ...
│       └── 460_noise.png
└── submissions/
    └── noisy/
        └── 461_noise.png
```



The denoised submission images must be saved inside `competition_data/submissions/denoised/`. Rename only the denoised outputs as follows:

461_noise.png	→	461.png
462_noise.png	→	462.png
⋮		
480_noise.png	→	480.png

- A baseline implementation is provided in the repository under `baseline/` as a reference example to demonstrate the expected workflow, directory structure, and input/output format. Teams are not required to use the baseline algorithm. They may improve the provided baseline or develop a completely different approach. The self-evaluation tool in `evaluation/evaluate.py` implements the same image-quality metrics and composite-score formula used for official scoring and may be used on ground-truth images available to the team.
- All custom scripts must be placed inside `scripts/`. Teams are not allowed to change files in other directories.
- Python must be used. External Python libraries are allowed.
- Teams may use a classical approach, a deep-learning approach, or a combination of both.

Hint: Google Colab or Kaggle may be used for GPU-based training.

Hint: The final script must include a CPU fallback. Test the code on a CPU using a few images to confirm that it runs within a reasonable time.

2.2 Evaluation Format

- Teams must create a Python file named `denoise.py` inside `scripts/`. Running this file must denoise the images in the given noisy-image directory and save the results in the given denoised-image directory. The provided `evaluation/evaluate.py` may be used separately for self-evaluation on image pairs for which ground truth is available.

Example command:

```
python scripts/denoise.py --noise_dir x --denoised_dir y
```

- Create a `README.md` file with clear instructions on how to run the script.
- The denoised outputs for images 461–480 must follow the renaming format given above. The script `scripts/denoise.py` must handle the renaming.
- Refer to `baseline/` for an example of how to organise the solution.
- The evaluation score is calculated as follows:

$$\Delta\text{PSNR} = \text{PSNR}_{\text{denoised}} - \text{PSNR}_{\text{noisy}}, \quad \Delta\text{SSIM} = \text{SSIM}_{\text{denoised}} - \text{SSIM}_{\text{noisy}}.$$

$$N = \text{clip}\left(\frac{\Delta\text{PSNR}}{15}, 0, 1\right), \quad S = \max(\Delta\text{SSIM}, 0).$$

$$\text{Evaluation Score} = 0.6N + 0.4S.$$

Teams may use `evaluation/evaluate.py` to evaluate their models on image pairs for which ground truth is available. The official evaluation uses the same PSNR, SSIM, improvement terms, and composite-score formula, together with additional organizer-side submission validation.

Baseline composite score: approximately 0.26

To be considered for ranking and shortlisting, a submission must outperform the provided baseline composite score. Teams may decide how to validate and improve their method using the given public dataset, as the validation strategy may depend on the selected architecture. These decisions must be clearly stated in the report.

2.3 Submission Format

- A team may submit multiple times, but only the last complete submission received before the deadline will be considered.
- Each team must submit one Google Drive folder containing the following two items:
 1. **TeamName.zip** — a ZIP file containing exactly the 20 denoised preliminary-round output images, named 461.png–480.png.
 2. **TeamName.Report.pdf** — the report describing the solution architecture, methodology, validation approach, and test results.
- The provided noisy images do not need to be included in TeamName.zip. The ZIP file should contain only the required 20 denoised output images unless the organizers specifically request otherwise.
- In addition to the Google Drive submission, each team must maintain its complete submitted solution in a private GitHub repository and add the official Mora SP Cup organizer GitHub account as a collaborator before the submission deadline. The repository should follow the provided starter-repository structure.
- The team must state the exact Git commit SHA of the submitted version in the repository README. The organizers will independently record this SHA at the submission deadline, and it will be treated as the frozen preliminary-round code submission.
- If trained model weights or other required binary files are too large to include directly in the repository, they may be uploaded separately to Google Drive. The link must be accessible to the organizers, and the team README must clearly state the model filename, expected local path, setup steps, and SHA-256 checksum. The organizers will independently verify and record the SHA-256 checksum of the submitted model file at the submission deadline. That exact model/checkpoint will be treated as frozen.

2.4 Competition Phases

Preliminary Round

August 29–September 19, 2026

Each team must submit its code together with a concise report explaining the design.

Teams must perform the denoising operation on the 20 images in `competition_data/submissions/noisy/` and save the output in `competition_data/submissions/denoised/`. These 20 outputs (images 461–480) will be evaluated against ground truth that remains hidden from participants. Images 481–500 are reserved for the physical final round and are not used during preliminary-round evaluation.

Preliminary Submission Package

Submit one Google Drive folder containing:

- TeamName.zip — exactly 461.png–480.png;
- TeamName.Report.pdf — the preliminary-round report.

The private GitHub repository URL and access to the official organizer GitHub account must also be provided as instructed by the organizers. If external model weights are required, the Google Drive model link must be documented in the repository README.

The report should include:

- A brief introduction to the problem.
- The problem and observations on the dataset.
- The solution architecture (diagrams are preferred).
- Justification of the selected approach.
- Alternatives that were considered.
- The methodology used to build the algorithm or model using the given 460 image pairs.
- A cover page is not required.
- Any other necessary information.

Important: The report must be kept under five pages, use a 12-point font, and have 1-inch margins on all sides. Keep the details to the point and present them with a clear flow, proper justification, and analysis. Failure to follow these requirements may lead to a penalty or disqualification.

Rename the report as `TeamName.Report.pdf`.

Codebase and Repository Requirements

- Create the `README.md` file inside `scripts/`, with clear instructions on how to run the code.
- If external Python libraries are used, include them in a requirements file.
- Create a `denoise.py` script that follows the required input/output format demonstrated by the provided baseline. Running the script must denoise the 20 input images and save the results in the given denoised-image directory. The internal algorithm or model may be completely different from the baseline.
- All inference must run locally on the test machine. The code must operate fully offline without making network requests once the required model files have been downloaded and placed in the documented location.
- Teams must add the official Mora SP Cup organizer GitHub account as a collaborator to the private GitHub repository containing the submitted code. The repository should preserve the provided starter-repository structure. The exact Git commit SHA used to generate the submitted outputs must be written in the repository `README` and will be recorded by the organizers at the submission deadline. **No new commits are permitted in the submitted repository after the submission deadline.** The recorded SHA must remain the latest submitted commit and will be used to verify the code before the final round.
- If a trained model or checkpoint is too large to include directly with the code, upload it to Google Drive and provide an organizer-accessible link. Clearly document the model filename, where it must be placed, how it is loaded, and its SHA-256 checksum in `scripts/README.md`. **The submitted model/checkpoint must not be replaced, modified, or re-uploaded after the submission deadline, even if the same Google Drive link is retained.** The organizers will independently verify and record the SHA-256 checksum and will use it to confirm that the same model is used during the final round.
- The preliminary image submission must be packaged as `TeamName.zip` and must contain exactly the denoised outputs `461.png`–`480.png`. The provided noisy images do not need to be re-submitted.
- The code must be clear and readable.
- Hardcoding or direct lookups will result in immediate disqualification.
- If the submitted code does not generate the submitted denoised images, the team will be disqualified.
- Hardcoded image-specific mappings, direct lookups, or other forms of dataset exploitation that do not represent a generalizable denoising procedure may result in a penalty or disqualification.

Important: Submission Freeze

- **Code freeze:** The Git commit SHA stated in the `README` and recorded by the organizers at the deadline is the official submitted code version. **No commits may be made to the submitted repository after the deadline.**
- **Model freeze:** The SHA-256 checksum of every submitted model/checkpoint will be recorded by the organizers. **Model files must not be modified or replaced after the deadline.**
- Before the final round, the organizers may verify that the latest GitHub commit SHA matches the recorded submission SHA and that the downloaded model file produces the same recorded SHA-256 checksum.
- Any mismatch or post-deadline modification to the submitted code or model may result in a penalty or disqualification.

Example submitted repository structure

```

baseline/
├── ...
└── competition_data/
    ├── submissions/
    │   ├── noisy/
    │   │   ├── 461.noise.png
    │   │   ├── ...
    │   │   └── 480.noise.png
    │   └── denoised/
    │       ├── 461.png
    │       ├── ...
    │       └── 480.png
    └── evaluation/
        ├── evaluate.py
        └── scripts/
            ├── README.md
            └── requirements.txt

```

```

├── denoise.py
├── (all other files required by the solution)
└── README.md

```

Note: Large trained model files do not need to be committed directly to the repository. If required, provide them through Google Drive and document the download link, expected local path, and SHA-256 checksum in the README. All required model files must be available to the organizers before the submission deadline. The submitted model file is frozen at the deadline and must not be replaced or modified afterward, even at the same Drive link.

Final Round

Exact dates will be announced later.

- This round will be held physically at the University of Moratuwa.
- Five teams will be selected based on their preliminary-round submissions. The selected teams must present their work and architecture, justify their design choices, and present the overall performance. A technical question-and-answer session and viva will be conducted to evaluate each team's understanding, originality, and implementation rigour.
- Teams selected for the final round must use the same code version and the same trained model/checkpoint submitted for the preliminary round. The organizers will verify the code using the recorded Git commit SHA and the model file using its recorded SHA-256 checksum. The submitted repository must contain no post-deadline commits, and the submitted model/checkpoint must not have been replaced or modified. Any mismatch may result in a penalty or disqualification.
- Each team will receive the 20 previously hidden final-round noisy images (images 481–500) and must perform the denoising task in front of the invigilators and judges using the same submitted code and model.
- The winners will be decided based on performance on the 20 new images, the presentation, and the viva.
- The judges' decision will be final.

3. Evaluation Metrics

Preliminary Round

- **40%** — Report and technical analysis.
- **10%** — Use of significant classical signal-processing techniques beyond the baseline, with proper justification.
- **35%** — Score received for the 20 preliminary-round submission images (461–480), evaluated against their hidden ground truth.
These points will not be awarded if an error occurs while running the submission. Make sure that the image formats are correct.
- **15%** — Clear code structure, adherence to the guidelines, and better runtime on a standard mid-range CPU.

4. Eligibility Criteria

- A team must have a minimum of one member and a maximum of three members.
- The competition is open to all currently registered undergraduate students from any state or private university in Sri Lanka.
- All members of a team must be from the same university.
- Each participant must submit valid proof of identity confirming that they are currently an undergraduate student who meets the above criteria.

Registration Link

<https://docs.google.com/forms/d/e/1FAIpQLSd2XjB2oNAFF7PnIDRXWwQ3xwV-LcWfP6uNBCjbSuIEFiFx3w/viewform>

Teams must register before September 2, 2026, at 11:59 p.m. local time.

5. Resources

- Google Colab — free T4 GPU access:
<https://colab.research.google.com/>
- Kaggle — 30 free GPU hours per week:
<https://www.kaggle.com/code>

Dataset Usage

The competition dataset is provided for participation in Mora SP Cup 2026 and related educational or research activities. Participants must not redistribute, publicly mirror, sell, or republish the dataset without permission from the organizing committee. Hidden ground-truth images and final-round data must not be obtained, reconstructed, or redistributed.

Key Libraries and Frameworks

- **PyTorch and Kornia:** PyTorch-based differentiable image processing.
- **OpenCV and Scikit-Image:** Python tools for classical bilateral filtering, non-local means, and wavelet denoising.

6. Contact Us

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